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Amoeba-Inspired Self-Healing Electronic Slime for Adaptable, Durable Epidermal Wearable Electronics

Yu Feng, Cong Wu, Meng Chen, Hui Sun, Arul Lenus Roy Vellaisamy, Walid A. Daoud, Xinge Yu, Guanglie Zhang,* and Wen Jung Li*

Epidermal electronics has garnered significant research attention due to its promising applications in wearable human-machine interfaces and intelligent healthcare sensing. However, their widespread use faces challenges due to complex manufacturing processes, high material costs, inadaptability to different skin surfaces, and inadequate reusability. Herein, inspired by the biological reshapability and environmental adaptability of amoeba, an ultra-deformable ($\approx 2600\%$ strain), bioadhesive (adhesive strength ≈ 3 kPa), strong self-healing (fastest recovery time ≈ 1 s, maximum wound distance ≈ 5 mm), and electromechanical-durable wearable electronic slime (E-slime) is proposed, which can instantaneously form on-skin electronics in situ to detect body motion and physiological signals. E-slime demonstrates desired sensing performance with high sensitivity (gauge factor 2.95), wide sensing range (up to 400% strain), and low detection limit ($\approx 1\%$ strain), which can seamlessly adhere to the skin and can be easily reused multiple times (≈ 100 cycles usage). E-slime also enables on-the-fly deployment of motion monitoring tasks at various body locations, showcasing its versatility and reliability for body motion recognition and personal health monitoring. This study holds potential for next-generation green electronics, motion sensing devices, and wearable human-machine interfaces, ultimately helping to ensure healthy lives and promote well-being.

1. Introduction

Epidermal electronics, a type of flexible electronic system, can be applied to the body surface to perform tasks such as monitoring physiological signals,^[1,2] facilitating human-machine interaction,^[3–5] and providing assistive therapy.^[6,7] This type of flexible electronic system can be classified into various forms, including electronic patches, electronic tattoos, and electronic ink, based on their physical characteristics.^[8–15] Despite recent advancements in creating thin and well-integrated electronic patches,^[16–19] they still face limitations regarding their inability to be customized in terms of shape and size for specific applications. Conversely, electronic tattoos offer a solution to this issue by pre-fabricating and tailoring the desired morphology prior to application on the skin.^[20–23] However, the complexity of the fabrication process and the requirement for transfer printing during deployment pose challenges to their conformability with the skin. On the other hand, electronic ink allows for on-demand

customization by directly writing onto the targeted epidermal area.^[24–26] Nevertheless, electronic ink typically has a limited shelf life, and exposure to sweat on the body surface can cause its performance deterioration, detachment, or even failure.

In recent years, “Silly Putty-like” material, as a flexible and deformable material, has emerged and been utilized in the applications of soft robotics,^[27,28] circuit reparation,^[29] and flexible electronics.^[30,31] This material exhibits enough self-healing and malleability due to its exceptional viscoelasticity, allowing it to be easily shaped and molded to meet specific requirements. “Silly Putty-like” hydrogel, due to its biocompatibility, has sparked a growing interest in its application within the field of wearable electronics.^[32–36] Capitalizing on inherent deformability and bioadhesive properties, this hydrogel can be flexibly applied to the skin surface without the requirement for printing equipment or auxiliary fixation methods. However, the fabrication process of available hydrogel-based “Silly Putty-like” material with good conductivity is quite intricate, and the conductive composites used are still confined to a few options, such as silver/gold nanowires/particles, conductive polymers, MXene, and liquid metals.^[37] These existing composites come with certain limitations, including high cost,^[38–41] limited shape customization,^[42–44] and difficult reusability,^[45–47] hindering the

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widespread utilization of such materials in the field of epidermal electronics. In addition, although the electrical performance of “Silly Putty-like” hydrogels can be improved by increasing the doping ratio of the conductive active materials to the substrate, their mechanical properties may be adversely affected. Hence, the development of “Silly Putty-like” hydrogels that combine appropriate electrical performance, mechanical performance, adhesive properties, and self-healing capability remains a challenging task. Consequently, “Silly Putty-like” hydrogel, with the ability to overcome these obstacles, holds great promise in achieving electromechanical accessibility, cost-effectiveness, easy customization, and long-term stability for epidermal electronics across a broad spectrum of practical applications.

Amoebas, as unicellular organisms capable of promptly altering their shape in response to external stimuli, exhibit remarkable adaptability to their surrounding environment. They move toward a cue on surfaces, accompanied by their change in shape because amoebas extend their cytoplasm and elastic plasmalemma outward through actin polymerization, which leads to the formation of temporary arm-like pseudopods that helps them to stick to the substratum for gaining traction and thus propel themselves forward (Figure 1a,b).^[48,49] Inspired by the biological plasticity and environmental adaptability of amoebas, herein, this work utilizes biocompatible polyvinyl alcohol (PVA) as a matrix and introduces flexible epidermal electronics called electronic slime (E-slime) through a simple three-step fabrication method involving tannic acid (TA), glycerol (gly), and cost-effective carbon-based conductive materials (graphite and carbon black). As shown in Figure 1c, E-slime, as a type of “Silly Putty-like” hydrogel, can undergo deformation in multiple different patterns on the skin due to its fully reusable and easily customized properties. Benefiting from the ultra-conformal ability, E-slime can instantly form electronic skin in situ on the targeted epidermal area tightly, even on the wrinkled skin (Figure 1d) due to the rigid anchor points constructed by inner crosslinking of E-slime (Figure 1e); then the signals associated with the skin motion can be obtained (Figure 1f). Furthermore, Due to its moderate elastic modulus (≈ 5.3 kPa) and solid strength (up to ≈ 2 kPa for ≈ 40 cycles), E-slime can be easily peeled off from the attached area, reshaped into a different form, and re-deployed for sensing in a new target area. E-slime exhibits the ability to withstand external mechanical disturbances due to its robust self-healing cycle (Figures S6 and S7, Supporting Information) and repeatable self-healing capability, expanding its range of applications under dynamic conditions. As illustrated in Figure 1g, E-slime can be conveniently deployed at different locations on the body surface to measure various human motion signals, including joint extension and flexion, throat movements, facial expressions, and pulse. These demonstrations further validate the multifunctionality, reliability, and applicability of E-slime in personal health monitoring and body motion recognition. Moreover, Figure 1h indicates that E-slime exhibits desired conductivity (0.33 S m^{-1}), gauge factor (2.95), sensing range (up to 400%), stretchability ($\approx 2600\%$), and self-healing time (≈ 1 s) in comparison to similar “Silly Putty-like” hydrogels offering comparable information. Additionally, more performance comparisons (detection limit $\approx 1\%$ strain, maximum self-healing distance up to ≈ 5 mm, and self-healing cycles ≈ 30 times) provided in Table S1 (Supporting Information for details) highlight E-slime’s holistic blend of appropriate

electromechanical properties, reusability, and resilient self-healing capacity. This work paves the way for the development of next-generation artificial electronic skin, green electronics, intelligent medical sensing devices, and wearable human-machine interfaces.

2. Results and Discussion

2.1. Schematic Fabrication, Crosslinking Structure, Conductive Mechanism, and Structural Characterizations of E-Slime

PVA, as a widely used synthetic polymer with biocompatibility, has been employed to make different kinds of hydrogels with various functions for biomedical applications and human-machine interfaces. Here, accompanied by easy-to-obtain, low-cost, and environmentally-friendly raw composites (Tables S2 and S3, Supporting Information), PVA is used as the matrix material in the preparation of E-slime in three steps, as depicted in Figure 2a. First, a mixed solution was prepared with PVA, glycerol, and TA (chemical structure in Figure S1, Supporting Information) in a desired concentration in an 80°C water bath for 2 h stirring. After the dissolution of all chemicals, graphite and carbon black powder (chemical structure in Figure S2, Supporting Information) were added to the mixture in a 60°C water bath. Although graphite and carbon black are not soluble in water, TA monomer is able to undergo oxidative polymerization to form deposition onto the surface of graphite, while carbon black is dispersed in the interlayer spacing due to the high reduction ability of modified graphite by TA, achieving uniform mixing. Then, a borax pentahydrate solution was added drop by drop into the conductive mixture with continuous stirring to generate an E-slime.

Figure 2b illustrates the molecular crosslinking process in the synthesis of E-slime. The PVA solution is dissolved with glycerol and TA, forming the hydrogen bonds between catechol groups, respectively. Graphite and carbon black, as conductive particles, are doped and evenly distributed into the intermediate solution to create a prepolymer solution. To obtain E-slime, borax pentahydrate plays a role as a chemical crosslinker, inducing the polymerization of the prepolymer mixture. As indicated in Figure 2c, the establishment of a hierarchical three-dimensional hydrogel network in E-slime is facilitated by the intertwining of polymer chains across various substances, the presence of dynamic borate ester bonds between different components, and the concurrent occurrence of intramolecular and intermolecular hydrogen bonds. To improve the conductivity of the material, carbon black granules are incorporated as a filler into the interlayer spacing of the graphite sheet to establish the island-bridge structure (Figure 2d). Carbon black has an amorphous, isotropic structure, allowing for uniform conductivity in all directions. Graphite, with its layered structure, has high conductivity within each layer but lower conductivity between layers. Consequently, the conductivity can be enhanced by combining graphite and carbon black in a suitable ratio, which helps to bridge the graphite planes with the carbon black particles. The amorphous hydrogel structure of E-slime was characterized by scanning electron microscopy (SEM), X-ray photoelectron spectroscopy (XPS), Fourier transform infrared spectroscopy (FTIR), and viscosity test. The microstructure of E-slime imaged by SEM indicates that the hydrogel is relatively compactly crosslinked with the establishment of

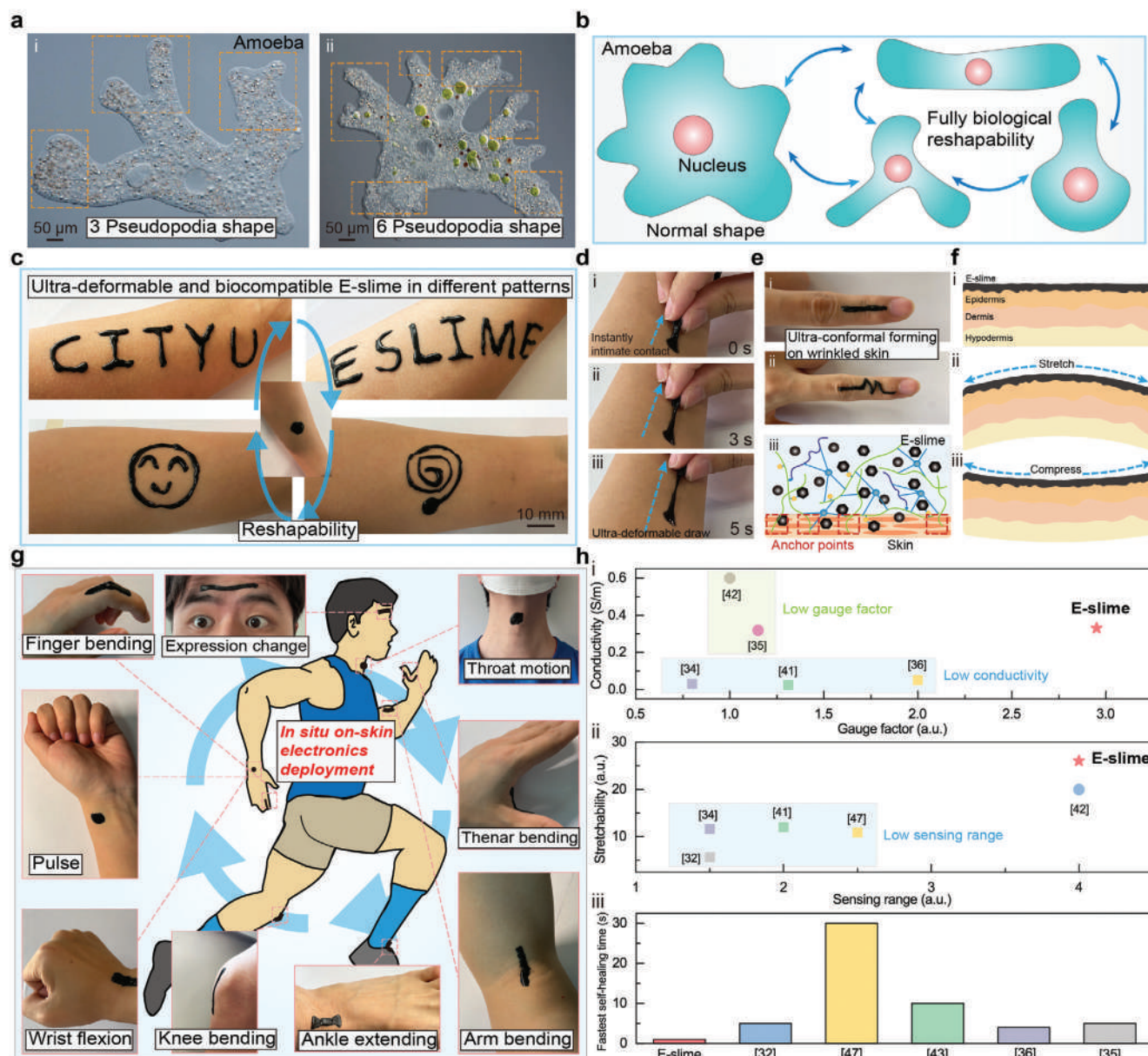


Figure 1. The properties and applications of amoeba-inspired E-slime. a) The optical photos (no copyright) of amoeba with shape-morphing pseudopodia. b) The full biological reshaping of amoeba from different morphologies. c) The easily customized and fully reusable properties of E-slime on skin. d) The instant draw of E-slime on skin with ultra-deformable properties. e) The ultra-conformal properties of E-slime on skin. f) Schematic of the E-slime on skin subject to stretch and compression. g) In situ on-skin electronics applications of E-slime for human motion detection. h) Comparison between this work and references of “Silly Putty-like” hydrogels in terms of gauge factor versus conductivity, sensing range versus stretchability, and self-healing time.

the island-bridge structure (graphite and carbon black), as shown in Figure 2e. XPS spectra show that all four hydrogels have similar peaks (Figure 2f), and the high-resolution XPS of O 1s and C 1s (Figure S3, Supporting Information) indicate the appearance of C–O, C–O–C, and C–C due to these materials’ formation (C, H, and O). C–O bonds suggest interactions between PVA and borate, indicating crosslinking, while C–O–C linkages confirm the integrity of PVA’s polymeric backbone. C–C bonds indicate the presence of stable carbon frameworks within the organic components of the hydrogels. FTIR spectroscopy illustrates the

interaction and structure formation among PVA, glycerol, TA, and borax (Figure 2g). The peak at 1333 cm^{-1} was assigned to the C–O stretching mode of the ester part and the –OH bond plane vibration, respectively. These peaks are typical absorption features of TA and indicate its successful combination with other components in E-slime. The peak at 1040 cm^{-1} was primarily attributed to C–O stretching vibrations, which can indicate the extent of cross-linking within glycerol and PVA of the hydrogel. The small peak at 1404 cm^{-1} is caused by the bending of the B–O–B bond in the borate network, confirming the presence of

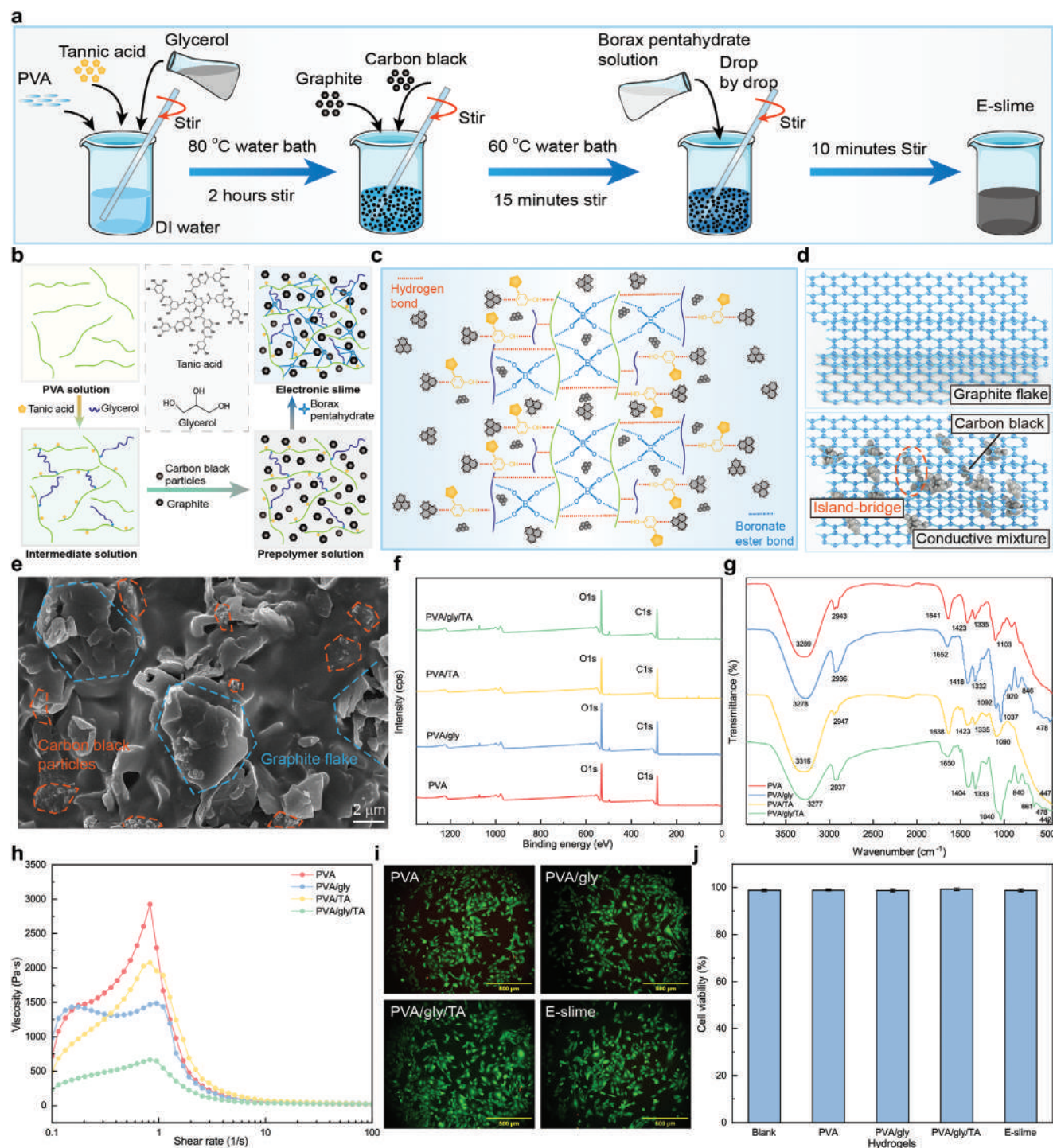


Figure 2. The schematics of fabrication, crosslinking structure, conductive mechanism, and characterizations of E-slime. a) The schematic preparation of E-slime. b) The crosslinking structure of E-slime. c) The bonding structure of E-slime. d) The conductive mechanism of E-slime. e) The SEM image of E-slime. f) The XPS spectra of prepared hydrogels (PVA, PVA/TA, PVA/gly, and PVA/gly/TA). g) The FTIR spectra of prepared hydrogels. h) The viscosity of the prepared hydrogels at room temperature. i) Fluorescence images of the cultured C2C12 cells under different hydrogels. j) The cell viability of the prepared hydrogels.

crosslinking between borax and the PVA chain, forming a boronic ester bond.

Additionally, Figure 2h shows that E-slime exhibits a low viscosity by comparing the viscosity curve of different hydrogels based on PVA, indicating E-slime's sticky and thick consistency at low shear rates. When the shear rate increases from 1 s^{-1} , E-slime illustrates a typical shear-thinning behavior owing to a rapid decrease of the viscosity from 665 Pa·s, verifying its property as a non-Newtonian hydrogel. The cell viability test of these hydrogels was conducted as shown in Figure 2i (experimental setup in Experimental Section). Figure 2j shows that the cell viability of E-slime is 98.72%, validating its biocompatibility. These characteristics demonstrate that E-slime is a highly flexible hydrogel, possessing a certain plasticity, showing the potential to be a desired candidate for epidermal electronics.

2.2. Mechanical and Electrical Characterizations of E-Slime

To investigate the glycerol's influence on the mechanical performance of PVA hydrogel, PVA hydrogels containing glycerol with different quality (6.5, 12.5, 17.5, and 22.5 wt%) were made for tensile experiments. As illustrated in Figure 3a, with the increase of glycerol content, the tensile strength of E-slime increases significantly from ≈ 60 to ≈ 110 kPa, while elongation at break decreases from $\approx 1800\%$ to $\approx 875\%$. In addition, the tensile strength and elongation at break have consistent changes when glycerol content increases from 17.5 to 22.5 wt%. The results suggest that as the amount of glycerol increases, the crosslinking density between different components also increases, resulting in the fracture of numerous hydrogen bonds to dissipate energy. Consequently, this leads to a decrease in the elongation at the breaking point of the hydrogel. Thus, PVA/gly hydrogel possesses the desired stretchability and mechanical strength with a mass concentration of 22.5 wt%. TA is a plant-derived polyphenolic substance that contains abundant catechol groups, which has gained widespread recognition as an excellent adhesive material. As shown in Figure 3b, the experiments were conducted to evaluate the mechanical effect of TA on PVA/gly hydrogel. With the increased content of TA from 0 to 15 wt%, the adhesive strength of PVA/gly/TA hydrogel increases from 2.42 to 5.84 kPa, while its elastic modulus decreases from 11.72 to 2.67 kPa. The results indicate that TA makes PVA/gly hydrogel more flexible and improves its adhesive ability. It is noteworthy that when the mass concentration of TA exceeds 15 wt%, the crosslinking process becomes unachievable, potentially due to the occurrence of aggregation between TA and PVA caused by an excessive amount of catechol moieties in TA. Thus, the mass concentration of TA is fixed at 10 wt% for subsequent research, where the hydrogel demonstrates robust adhesive strength (7.07 kPa) and possesses a suitable elastic modulus (5.45 kPa). To investigate the formal ratio of conductive particles (graphite and carbon black) to PVA/gly/TA hydrogel, the elastic modulus and conductivity of the prepared testing samples were evaluated, as illustrated in Figure 3c. With the increase of conductive dopant content, the elastic modulus and conductivity of E-slime increase, respectively. At the mass ratio of 1:1 (PVA/gly/TA hydrogel: conductive particles), E-slime has the appropriate elastic modulus (5.65 kPa), and the conductivity (0.31 S m^{-1}) has a sharp increase from a mass ratio

of 1:0.25 to 1:1 while a slow upward trend when the mass ratio increases from 1:1 to 1:1.5. Based on the above analysis, the mass ratio of 1:1 is chosen to fabricate E-slime for following research.

Figure 3d illustrates the sensing mechanism of E-slime. Carbon black particles act as a filler to create an island-bridge structure through conductive paths, enhancing the composites' overall conductivity by bridging the gaps between graphite layers and different graphite. When E-slime changes its shape, the internal structure of carbon-based composites will also be impacted, resulting in enhancement or diminishment of the conductive paths due to the change of composite location, consequently altering the electrical resistance of E-slime. Furthermore, to evaluate the effect of the carbon material composition ratio on the conductivity of E-slime, the relative resistance variation of E-slime was characterized by different mass ratios of graphite to carbon black. As shown in Figure 3e, the relative resistance variation changes positively with the increase of applied strain due to the effect of the elongated internal electron transfer channels. Moreover, with the rise of the mass ratio (graphite: carbon black), the average gauge factor of corresponding E-slime exhibits a trend of decreasing first and then increasing (Figure S4, Supporting Information). The highest gauge factor (≈ 2.95) occurs at the mass ratio of 2.5:1; continuing to expand the graphite content will reduce the connectivity of the carbon black, hindering the sensitivity, which is consistent with the aforementioned theoretical analysis. In addition, the sheet resistance with different thicknesses of E-slime (different mass ratios of graphite: carbon black) is further investigated (Figure 3f). Similarly, the sheet resistance presents a decreasing and then increasing trend with the mass ratio's increase no matter what thickness E-slime has, which is in accordance with the results of the gauge factor. Therefore, the ratio of graphite to carbon black is selected as 2.5:1 as the optimal ratio for E-slime in the subsequent study.

Figure 3g shows the strain resolution of E-slime under different strain loads (0% and 100% strain), illustrating that each strain increment correspondingly triggers a noticeable resistance change even with different reference strains. The electrical stability is a crucial parameter for the sensor made by hydrogel, which was investigated by evaluating the resistance change over time (room temperature), as shown in Figure 3h. All test samples (four different thicknesses) exhibit a resistance variation of $<5\%$ within the first 4 h, gradually increasing to $\approx 10\%$ until the 10th h. This result indicates that the sensor possesses essential electrical performance stability, making it suitable for relatively long-term storage and use. To further examine the durability of the electrical properties of the E-slime under mechanical stress, an analysis of the resistance of E-slime applied onto porcine skin (formation of in situ epidermal electronics) was conducted. The pigskin was subjected to various forms of deformation, including stretching ($\approx 20\%$), compression ($\approx 60^\circ$), contact, and twisting ($\approx 50^\circ$), to assess its stability. As depicted in Figure 3i, apart from the resistance increase ($\approx 19\%$) observed during the stretching process and the resistance decrease ($\approx 50\%$) during the contact process, the resistance remained consistently $\approx 2000 \Omega$. These findings indicate that the E-slime maintains strong adhesion and stable electrical conductivity, even when applied to an irregular and wrinkled skin surface and in the presence of different types of deformation.

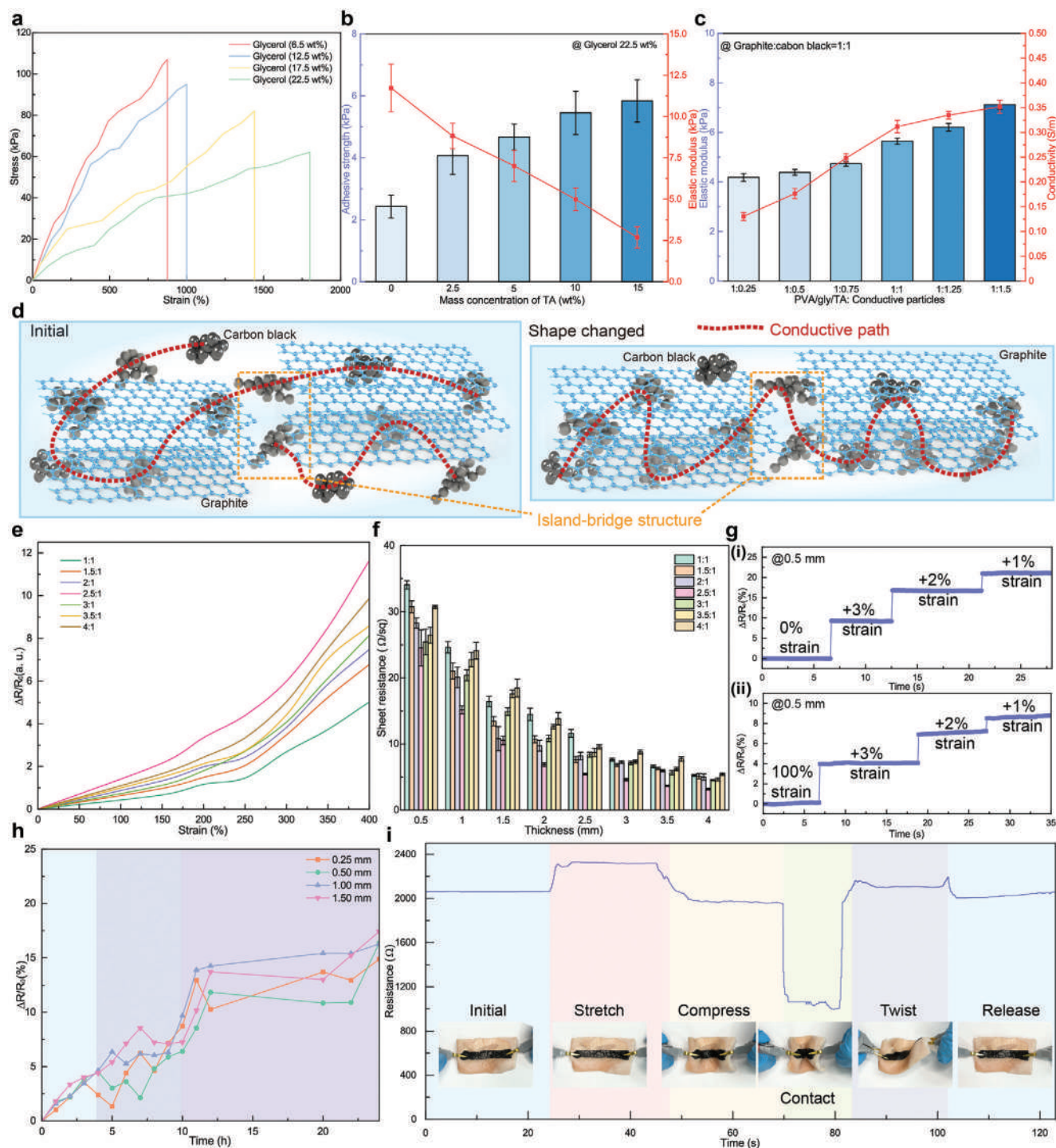


Figure 3. Mechanical and electrical characterizations of E-slime. a) The tensile stress-strain curves of PVA hydrogels with different mass content of glycerol. b) The adhesive strength and elastic modulus of PVA/gly hydrogels with different mass content of tannic acid. c) The elastic modulus and conductivity of PVA/gly/TA hydrogels with various mass ratios of conductive particles (mass ratio of graphite: carbon black = 1:1) to PVA. d) The conductive mechanism of E-slime under shape-changing. e) The relative resistance variation curves of E-slime as a function of strain with different graphite-to-carbon black mass ratios. f) The sheet resistance of E-slime under different thicknesses with various graphite-to-carbon black mass ratios. g) The strain resolution of E-slime under 0 and 100% strain load. h) The relative resistance variation of E-slime over time (room temperature) with different thicknesses. i) The resistance response of E-slime under different mechanical deformation on porcine skin.

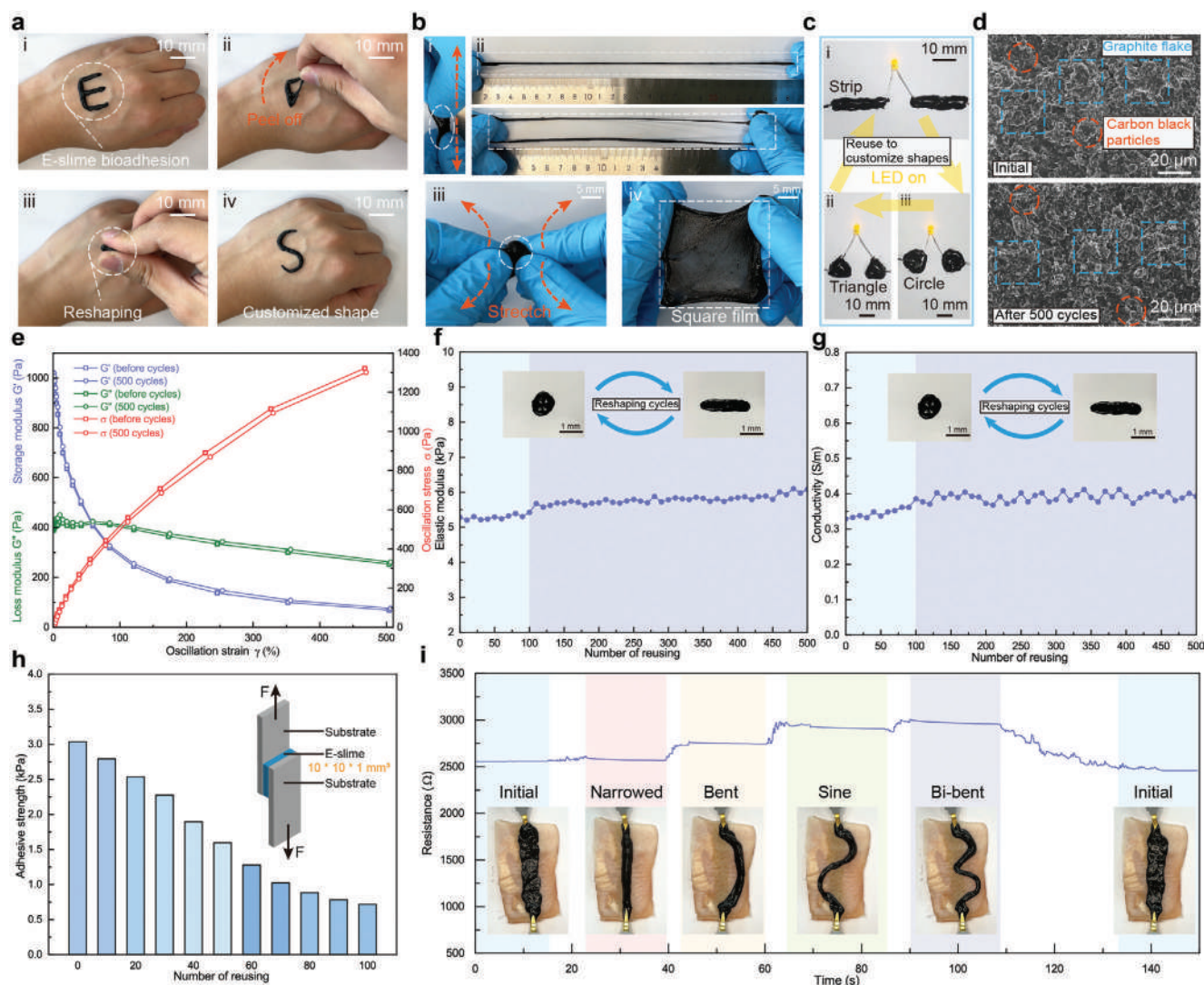


Figure 4. The easy customization and full reusability of E-slime a) The demonstration of E-slime reshaping for customized deployment on skin. b) The demonstration of E-slime that changes shape from ultra-long line to square film. c) The demonstration of E-slime's conductivity with reused and customized shapes. d) The SEM images of E-slime under different conditions (no-use and 500 cycles). e) The rheological characterizations of E-slime under different conditions (no-use and 500 cycles). f) The elastic modulus of E-slime under different usage times. g) The conductivity of E-slime under different usage times. h) The adhesive strength of E-slime under different usage times. i) The resistance response of E-slime when changing different shapes on porcine skin.

2.3. Easy Customization and Full Reusability of E-Slime

Figure 4a illustrates the procedure of E-slime's deployment and utilization for on-skin electronics. E-slime is easy to apply and adhere to the skin due to its bioadhesive ability, while it is also effortless to peel off, reshape, and reapply onto the skin in a demanded shape (more shapes and patterns in Figure S5, Supporting Information). Additionally, the impressive tensile strength of E-slime was demonstrated in the stretching test (up to $\approx 2600\%$ stretching without fracturing), as depicted in Figure 4b. Its notable mechanical properties also enable E-slime to form a thin film (thickness $\approx 400\ \mu\text{m}$), making it advantageous for its utilization as a strain sensor. As presented in Figure 4c, a piece of E-slime is connected to illuminate LED bulbs successively in dif-

ferent shapes (strip, triangle, and circle), demonstrating its electrical conductivity with full reusability and easy customization. The result validates that E-slime is a suitable conductive material for emergency circuit repairs and flexible electronic devices (Figures S8 and S9, Supporting Information). To confirm that E-slime can maintain its microstructure, the SEM images (initial state without use and 500 cycle uses) were obtained, as shown in Figure 4d. The graphite flakes and carbon black granules remain dispersed homogeneously in the crosslinked networks after repeated use 500 times, indicating E-slime's reusable stability. Furthermore, the rheological tests were conducted by evaluating the loss modulus, storage modulus, and oscillation stress before and after 500 times use (Figure 4e). At 250% oscillation strain, the loss modulus ($\approx 333\ \text{Pa}$) is apparently more extensive than the

corresponding storage modulus (≈ 137 Pa), whether before or after use. Moreover, before and after use, the curves of storage modulus and loss modulus remain almost unchanged, respectively. The results suggest that the E-slime is elastic and soft even after continuous cycled usage, which guarantees its easily customized capability.

Furthermore, the tests of E-slime's reusable electromechanical performance stability were conducted. Figure 4f presents the elastic modulus of E-slime under different usage cycles, showing that the elastic modulus gradually increases from 5.29 to 6.09 kPa after repeated use while remaining at ≈ 5.3 kPa under 100 cycles. The results suggest that the mechanical performance of E-slime will remain stable (variation $< 2.5\%$) with < 100 uses. Additionally, the electrical stability of E-slime is evaluated in Figure 4g, exhibiting a slow climbing trend (conductivity increases from 0.33 to 0.39 S m^{-1}) < 100 cycles, then a fluctuation (≈ 0.26 S m^{-1}) after 100 cycles. The findings indicate that the electrical performance of E-slime will also remain stable with < 100 uses. In addition to the electromechanical stability for reusable cycles, the reusable adhesion is also assessed, as shown in Figure 4h. The adhesive strength of E-slime progressively decreases during the using cycles from 3.03 kPa (initial) to 0.72 kPa (at 100 cycles), indicating that although repeated use can reduce the adhesiveness of E-slime, the adhesive strength still remains at a permissible level suitable for skin adhesion.

To further verify the easy customization and full reusability of the E-slime under various postures by in situ formation, an analysis of the resistance of the E-slime applied onto porcine skin was conducted. E-slime was subjected to multiple forms of deformation, including narrowing, bending, sine, and bi-bending, to evaluate its immediate strain response. As illustrated in Figure 4i, E-slime will obtain the corresponding different resistance response obviously during the whole shape-changing process and will receive the same response as the initial state after reshaping itself back. The aforementioned experimental results collectively demonstrate that E-slime exhibits suitable mechanical and electrical performance stability for repeated use (up to 100 times) while maintaining adequate bonding strength for repeated adhesion to wrinkled skin. This confirmed repeatability validates the convenient customization and full reusability of E-slime, further substantiating its viability as an in-situ deployable electronic skin.

2.4. Adhesive and Self-Healing Properties of E-Slime

Mussels are able to adhere to rocks and other uneven surfaces due to the presence of specialized proteins (containing catechol groups) in mussel foot secretions.^[50] Taking inspiration from the adhesive properties of mussels, E-slime demonstrates remarkable strength and durability in adhesion. This can be attributed to the abundant presence of catechol groups within the hydrogel matrix from TA, enabling robust and long-lasting adhesiveness on human skin. Figure 5a depicts that E-slime is able to adhere firmly to various substrates, including paper, glass, plastic, rubber, copper, iron, wood, fiber, and human skin, exhibiting a solid adhesion. In this process of adhesion for skin, the adhesive force is generated through the formation of two kinds of bonds. One

is the electrostatic interaction formed by borax pentahydrate and the collagens with abundant charged sites in the tissue; the other is the dynamic covalent bonds facilitated by TA in the adhesive layer and the amino or thiol groups in the tissue (Figure 5b). To evaluate the adhesive ability of E-slime, the adhesive strength was quantified by the tensile adhesive test. As shown in Figure 5c, E-slime exhibits the greatest adhesion strength of 7.91 kPa to paper while also offering a desired adhesive strength of 2.83 kPa to pigskin. The results allow E-slime to adhere firmly to the skin of the human body when monitoring human motion without any fixation.

As shown in Figure 5d, E-slime could conduct the circuit again and light up the LED light after self-healing. The self-healing process of E-slime is elucidated in Figure 5e, highlighting the underlying mechanisms. This process relies on two types of dynamic bonds. First, the healing is facilitated by the formation of dynamic catechol-borate bonds involving TA, enabling the reconstruction of the crosslinked network. Second, supramolecular interactions among PVA, glycerol, and TA molecules contribute to reversible bonds that are crucial for the healing processes. To investigate the self-repairing electrical performance of E-slime, it was connected to a real-time current measuring system for further analysis. Figure 5f clearly demonstrates that the current almost disappears when E-slime is completely severed. However, when the separated parts of E-slime are reconnected, the circuit immediately restores its original current value. This observation indicates the rapid reconstruction of the 3D conductive network among the contacted interfaces of the healed hydrogel. Remarkably, E-slime exhibits the ability to recover its original conductivity after severance in ten consecutive cut-healing cycles. Additionally, the relationship between separate distance and self-healing time was studied (Figure 5g). E-slime has a fast self-healing time (< 15 s) when the separate distance is < 2 mm, and it can still heal itself even when the distance increases to 5 mm. The results confirm that E-slime has a strong self-healing ability, enhancing its potential to resist external interference and damage in epidermal electronic applications.

Then, the influence of self-healing on the conductivity of E-slime is further investigated. As depicted in Figure 5h, the conductivity of E-slime will stabilize at its initial value (≈ 0.32 S m^{-1}) after 10 min, indicating that the electrical performance of the sensor remains unchanged before and after self-healing. Furthermore, the cut-healing cycle experiments were conducted to assess the robustness of the mechanical properties of E-slime. As illustrated in Figure 5i, the elastic modulus of E-slime is almost unchanged (≈ 5.5 kPa) even after 100 cut-healing cycles, verifying a robust mechanical property. At the same time, Figure 5j illustrates that E-slime's resistance variation was $< 5\%$ during the first 30 self-healing cycles. Even after 100 self-healing cycles, the relative resistance variation of the sensor remained below 7.5%. These results indicate that the sensor also possesses stable electrical self-healing performance. These findings validate the robust self-healing capabilities of the sensor, as well as the sustained stability of its electromechanical performance even after multiple self-healing cycles. This combination of electrical conductivity, self-recovery capability, and dynamic shaping ability positions E-slime as an ideal conductive material for in-situ soft electronics, particularly for applications in human motion detection.

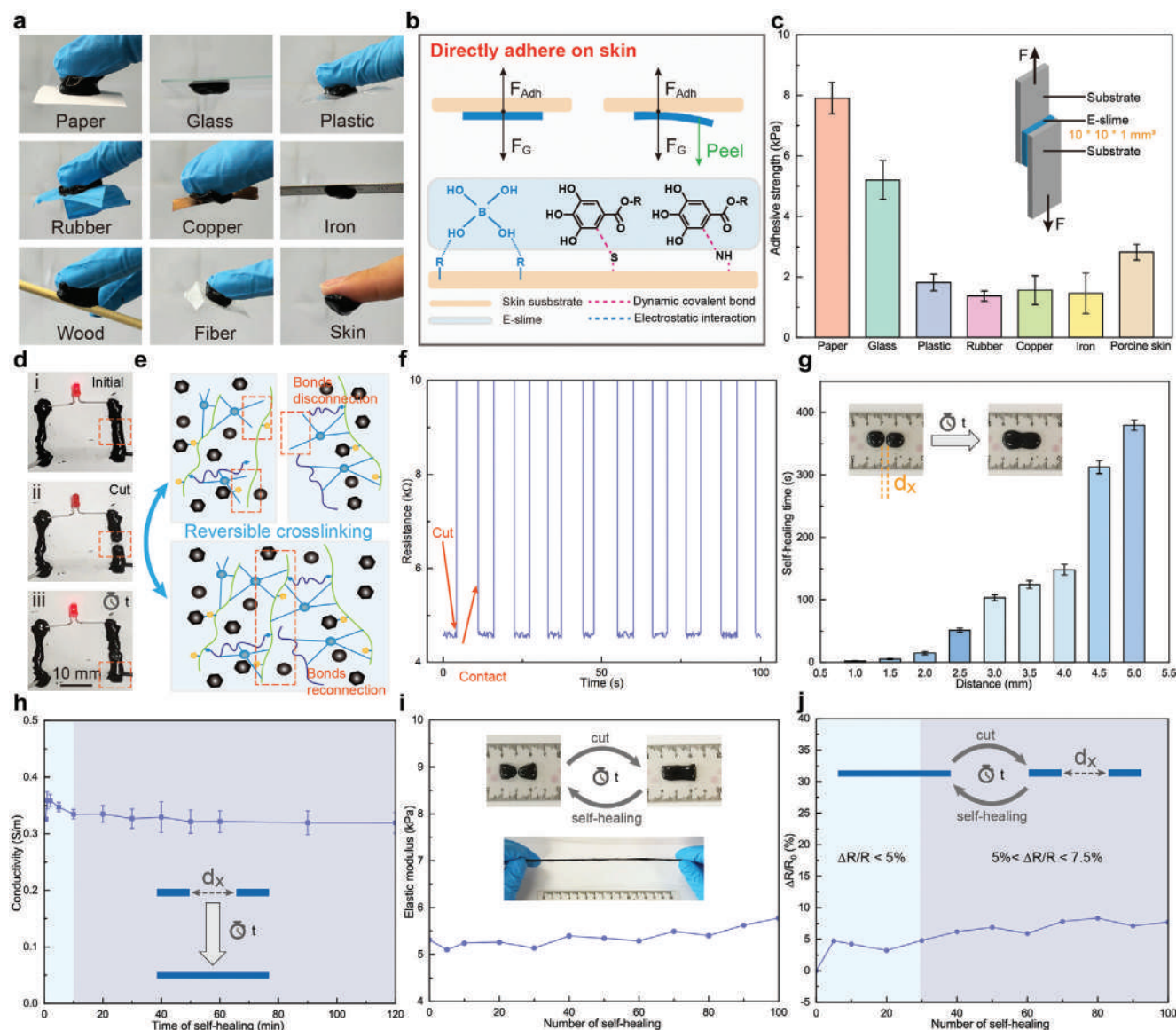


Figure 5. The adhesive and self-healing properties of E-slime. a) The demonstration of E-slime adhesive ability for various substrates. b) The principle of E-slime's bioadhesive property for skin. c) The characterization of E-slime's adhesive ability for different substrates. d) The demonstration of E-slime's self-healing property. e) The molecular mechanism of E-slime's self-healing property. f) The resistance change of E-slime during cutting–healing cycles. g) The self-healing period of E-slime under different cutting distances. h) The conductivity of E-slime under different cutting–healing cycles. i) The elastic modulus of E-slime under different cutting–healing cycles. j) The relative resistance variation of E-slime under different cutting–healing cycles.

2.5. Applications of E-Slime on Human Hand Motion Detection

Due to its appropriate electromechanical properties, plasticity, adhesion, and self-healing capabilities, E-slime holds significant potential as an excellent strain sensor. Moreover, its reusability and customizability enable it to be easily and repeatedly attached to various locations on the human body. Herein, the hand motion sensing ability of E-slime as a strain sensor was explored. **Figure 6a** displays an overview of mounting one piece of E-slime on various hand locations in different shapes to accommodate the corresponding motions, including the index finger (the first and second phalanges), thumb, thenar, and metacarpophalangeal joint (MCP joint).

The e-slime was directly attached to the finger (the second phalange) surface without any adhesive to monitor the dynamic movement of this joint angle (experimental setup in Figure S10, Supporting Information). As shown in Figure 6b, when the stretching angle of the finger increased from 60° to 120°, the relative resistance variation signal exhibited a correspondingly near-linear ($R^2 = 0.98$) decrease (sensitivity $\approx 0.17\%/^\circ$) due to the change of E-slime strain, indicating that E-slime has a high sensitivity. Figure 6c illustrates the electrical signal response of E-slime upon repeated measurements of finger bending motion at different angles (30°, 60°, and 120°). The results indicate that E-slime has a rapid response rate to finger bending motion (≈ 2 s) and the effectiveness of repeated motion detection; the results regarding

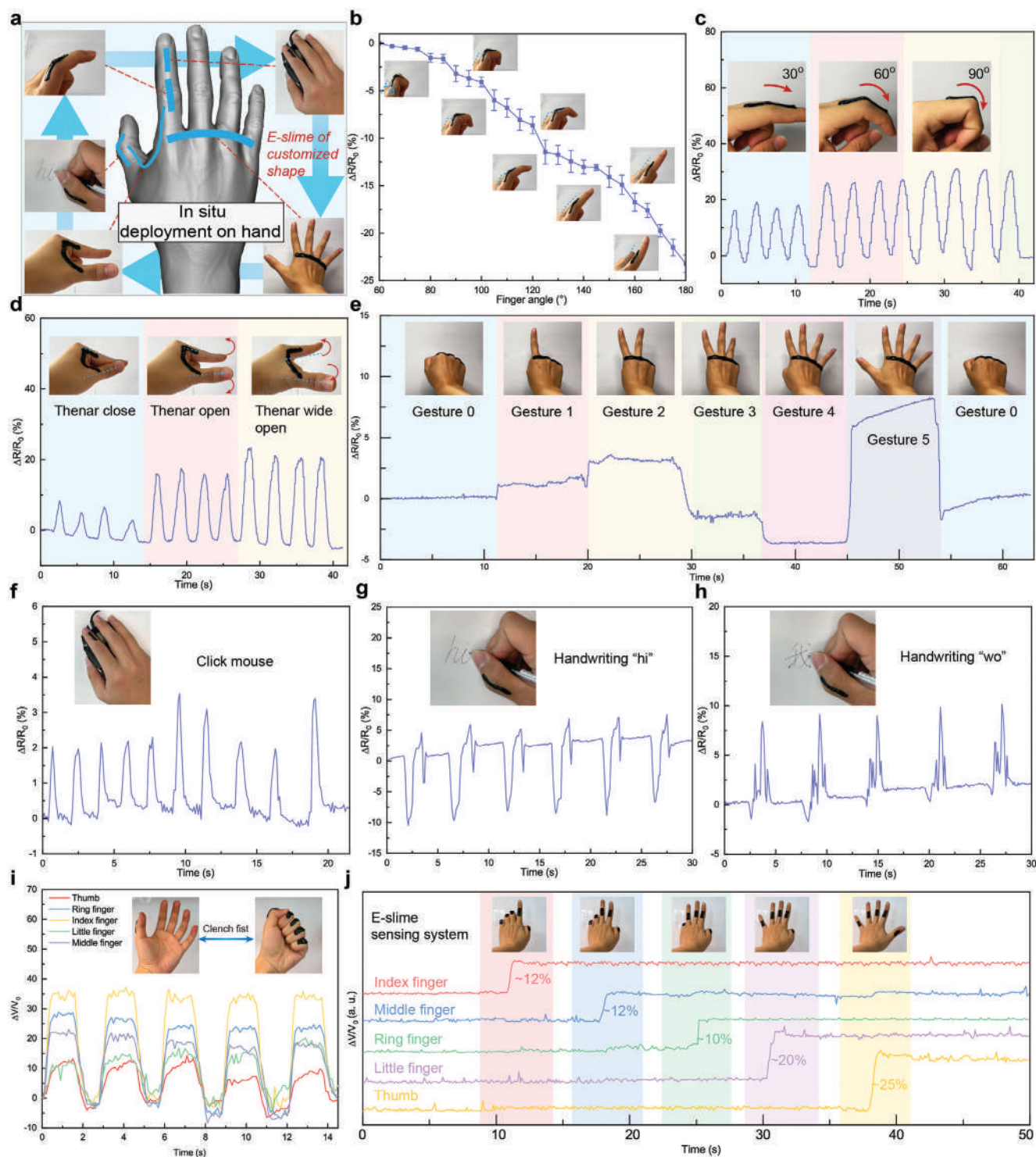


Figure 6. The applications of E-slime on human hand motion detection. a) The overview of sensing locations of E-slime placed on hand. b) The real-time E-slime's relative resistance variation as a function of finger position angle. c) The real-time relative resistance variation under different finger bending angles (30°, 60°, and 90°). d) The real-time relative resistance variation under different thenar motion positions. e) The real-time relative resistance variation under different hand gestures. f) The real-time relative resistance variation when clicking the mouse. g) The real-time relative resistance variation when writing "hi." h) The real-time relative resistance variation when writing the Chinese character "wo." i) The real-time relative resistance variation (five channels) when clenching fist. j) The real-time relative resistance variation (five channels) under different hand gestures.

cyclic detection (1200 cycles in ≈ 1500 s) of finger motion further verify the durability of E-slime (Figure S11, Supporting Information). As illustrated in Figure 6d, when E-slime was mounted in hand thenar, clear and stable signals of relative resistance variation could be detected as the generation of close and open (two angles) motion. To further investigate the application of E-slime for entire hand motion detection, it was adhered to the MCP joint. As illustrated in Figure 6e, each gesture, including “zero,” “one,” “two,” “three,” “four,” and “five,” produced a distinct resistance value that could be fully recovered.

In addition to the application for hand motion detection, E-slime is also capable of monitoring the subtle movement of the hand due to its sensing abilities of motion speed and direction aforementioned. By deploying E-slime onto the index finger, as demonstrated in Figure 6f, the signal of relative resistance variation ($\approx 2.5\%$) could be captured clearly when clicking the mouse each time. Although handwriting detection is challenging due to the involvement of unique characteristics (force, speed, and sequence of the writing), E-slime could produce a complex but stable waveform for even a simple signature, as shown in Figure 6g,h. The results demonstrate that E-slime is able to detect not only English words but also Chinese characters, manifesting the E-slime’s potential in the application of real-time handwriting recognition. Besides, an E-slime sensing system could be integrated through multiple pieces of E-slime units. As illustrated in Figure 6i–j, the E-slime sensing system (five sensing units) was able to detect five finger motion signals simultaneously, including the cycles of the clenching fist (five cycles) and hand gestures (five types). The aforementioned experimental results validate the application of E-slime in hand motion posture detection, further highlighting its potential as a flexible sensor or epidermal sensing system that can be formed in situ on the hand. This emphasizes its potential in applications for human-computer interfaces and gesture pattern recognition.

2.6. Applications of E-Slime on Human Body Motion Detection

Beneficial from the convenient customizability, full reusability, strong skin adhesion, stable self-healing capabilities, and adaptability to the skin, E-slime can be applied to various areas of the human body for detecting motion signals by in-situ formation of epidermal electronics, including joint movements and physiological micro-motions (Figure 7a). As demonstrated in Figure 7b, E-slime effectively detects facial protrusions and captures signals during frowning (relative resistance variation $\approx -3\%$) and expresses amazement (relative resistance variation $\approx 7\%$) when attaching to the forehead. Additionally, when E-slime is adequately adhered to the side of the mouth, the signals caused by smiling (relative resistance variation $\approx 12\%$), laughing (relative resistance variation $\approx 2\%$), and even big laughter (relative resistance variation $\approx 5\%$) can be consistently and clearly identified (Figure 7c), demonstrating the potential of E-slime in facial expression recognition applications.

Besides facial motion detection, E-slime can also be applied as a flexible wearable sensor for body joint motion detection. As illustrated in Figure 7d, E-slime could accurately measure the electrical signals when the wrist is bent by 45° multiple times in both clockwise (relative resistance variation $\approx -25\%$) and counter-

clockwise (relative resistance variation $\approx 10\%$) directions. When the elbow was bent multiple times at angles of 30° , 45° , and 60° , E-slime was capable of measuring the corresponding relative resistance changes ($\approx 11\%$, $\approx 19\%$, and 25%) clearly and consistently (Figure 7e). Furthermore, as shown in Figure 7f, E-slime could also continuously detect the signal changes during multiple knee flexion and extension movements at different angles, including 30° flexion (relative resistance variation $\approx 2\%$), 30° extension (relative resistance variation $\approx -10\%$), and 60° extension (relative resistance variation $\approx -15\%$). As demonstrated in Figure 7g, when altering the shape and size of E-slime and attaching it to the ankle joint, it could accurately detect the electrical signal changes during joint extension (relative resistance variation -15%) and flexion (relative resistance variation $\approx 20\%$). The above results indicate that changing the deployment position and shape dimensions of E-slime can accurately and distinctly detect the electrical signals of different body movements in a targeted manner. This further demonstrates its potential for human-computer interaction applications.

Moreover, the human body surface manifests delicate mechanical signals, including strain in the throat area triggered by vocalizations and strain in pulse locations. Thanks to E-slime’s exceptional sensitivity and ease of customization (in terms of shape and thickness), it is capable of detecting these faint signals. As demonstrated in Figure 7h, E-slime was specifically tailored into a thin rectangular film form and applied to the skin in the throat region, exhibiting a high sensitivity in detecting the corresponding electrical signals when swallowing (relative resistance variation $\approx 15\%$) or coughing (relative resistance variation $\approx 8\%$) occurs. Additionally, when E-slime is customized into a circular disc shape and placed on the wrist at the pulse location, it can consistently and accurately detect the pulse (≈ 72 bpm) of the minute strains caused by pulse signals (Figure 7i). In conclusion, the results confirm that E-slime exhibits excellent sensitivity, electromechanical stability, robust bioadhesion, reliable self-healing capability, and the added advantages of reusability and easy customization. It can accurately detect both extensive human body motions (large deformations) and subtle changes in epidermal motion. As a result, E-slime emerges as a promising wearable strain sensor with flexible properties, offering significant potential for applications in human-computer interaction, facial expression recognition, and health monitoring.

3. Conclusion

Epidermal electronics play an important role in healthcare and human-machine interfaces due to their ability to provide biological signal monitoring, detect body movements, and sense subtle motions through custom designs tailored to individual body shapes and sizes. However, widespread utilization of epidermal electronics encounters barriers related to complex production methods and high material costs. Developing epidermal electronics that achieve appropriate electrical and mechanical performance, adhesive properties, and self-healing capability remains an ongoing challenge. In this work, an ultra-deformable, bioadhesive, self-healing, and electromechanical-durable wearable electronic slime for epidermal electronics applications inspired by the shapeshifting abilities of amoeba is developed. E-slime is based on a biocompatible PVA/gly/TA hydrogel

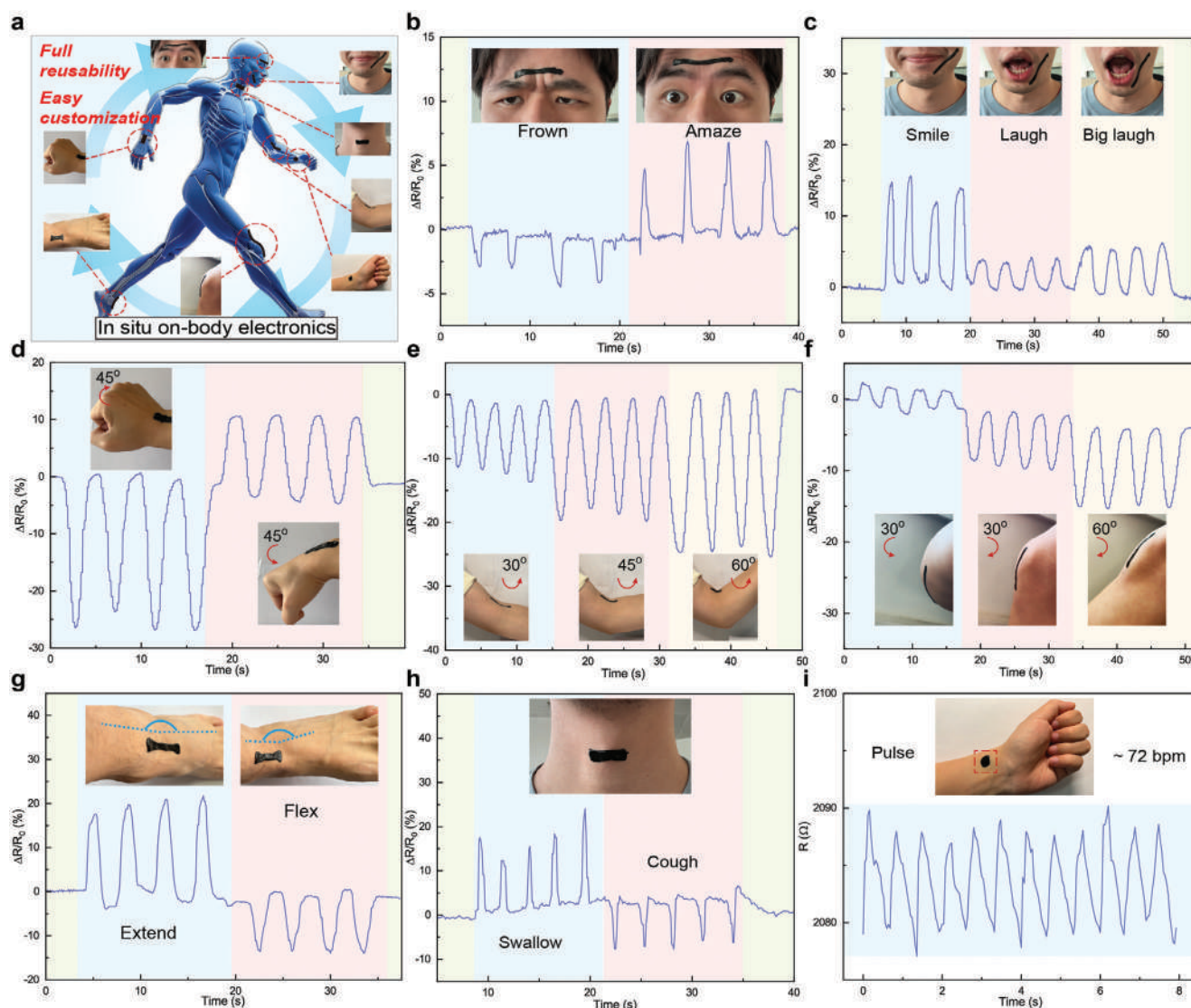


Figure 7. The applications of E-slime on human body motion detection. a) The overview of sensing locations of E-slime applied on the body. b) The real-time relative resistance variation under different facial expressions (frown and amaze). c) The real-time relative resistance variation under different mouth postures (smile, mouth open, and wide mouth open). d) The real-time relative resistance variation under different wrist bending angles (45° clockwise and counterclockwise). e) The real-time relative resistance variation under different elbow bending angles (30°, 45°, and 60°). f) The real-time relative resistance variation under different knee bending angles (45° clockwise/counterclockwise and 60° clockwise). g) The real-time relative resistance variation regarding ankle extension and flexion. h) The real-time relative resistance variation when swallowing and coughing. i) The real-time relative resistance variation when detecting pulse.

formulation, and it can be easily fabricated via a simple water bath process using raw materials that are readily available, low-cost, and environmentally friendly. Through the introduction of conductive carbon-based dopants of graphite and carbon black, E-slime achieves the desired electrical, mechanical, and physiological sensing properties for flexible electronics. Furthermore, E-slime also exhibits high conformability, adhesiveness, and self-healing attributes that allow it to seamlessly interface with skin and withstand external mechanical interferences. In addition, it can be reused multiple times through simple peeling-off and reapplication processes. Thus, E-slime has been applied to motion detection across diverse human body locations in this work, and the experimental results demon-

strate the versatility of E-slime, which has great potential for applications in personal health tracking and human-machine interfacing.

4. Experimental Section

Materials: Polyvinyl alcohol (PVA) with a molecular weight of 89 000–98 000, tannic acid (TA) and glycerol (gly) with analytical reagent, borax pentahydrate with a 100 mesh, and graphite powder with a 10 000 mesh were bought from Shanghai Macklin Biochemical Technology Co., Ltd. Carbon black powder with 200 mesh was supplied by Lion Specialty Chemicals Co., Ltd. All chemicals, solvents, and other consumable materials were purchased and used as received unless specially noted.

Preparation of E-Slime: Under constant stirring with a water bath at 80 °C for 2 h, 0.35 g PVA, 0.1 g gly, and 0.075 g TA were added and dispersed into 10 g deionized water. Then, 0.25 g graphite and 0.1 g carbon black were dissolved into the mixture and stirred for 10 min for complete dissolution under a 60 °C water bath. After stirring, 4% borax pentahydrate solution was added to the mixture drop by drop under stirring for polymerization. Finally, the electronic slime was obtained after 10 min of stirring.

Skin Replica Fabrication: Except for directly measuring the human electrical signal, pigskin purchased from a grocery store and cut into small pieces was utilized as a surrogate for human skin in all the experiments conducted in this paper. To emulate human skin, the pigskin undergoes degreasing and hair removal.

Tensile Tests of E-Slime: The tensile test was evaluated on a tensimeter (LT-5000, Reotai Precision Instrument Co., Ltd) equipped with a 10 N load cell at the crosshead speed of 60 mm min⁻¹. The tested cylindrical samples were 20 mm long and with a gauge length of 5 mm.

Adhesion Performance Tests of E-Slime: The adhesion strength tests were carried out on the tensimeter (LT-5000, Reotai Precision Instrument Co., Ltd). The tested sample was stuck to two pieces of substrates with a bonding area of 10 × 10 mm². After the sample bonded together, the adhered substrates were conducted to a lap shear stress when applied to the aforementioned tensimeter with a crosshead speed of 10 mm min⁻¹. By calculating the maximum force divided by the initial adhered area, the maximum stress was obtained as the adhesive strength.

Reusability Tests of E-Slime: A testing sample of E-slime changed shape from circular to rectangular, which was repeated as a process for each cycle. For each reusing cycle, the elastic modulus, conductivity, and adhesive strength were tested.

Self-healing Performance Tests of E-Slime: A testing sample of E-slime was cut into two pieces with a distance, which was repeated as a process for each cycle. For each self-healing cycle, the elastic modulus and conductivity were tested.

Electrical Tests of E-Slime: E-slime was prepared into a cubic shape with a size of 20 mm length, 10 mm width, and 2 mm thickness for the electrical performance tests. By using a digital multimeter (TH1963A, Tonghui Electronics Co., Ltd), the electrical signals of E-slime were obtained. The conductivity (σ) of the E-slime testing sample was calculated by the formula $\sigma = L/(R \times A)$, where L is the length of the testing sample, R is the resistance of the sample, and A represents the cross-sectional area of the sample. As for the relative resistance variation, it was determined by the equation $\Delta R/R_0 = (R - R_0)/R_0$, where R_0 and R are the resistance without and with applied strain, respectively.

Experimental Setup of E-Slime on Motion Detection: E-slime was applied to the targeted location of the human body in different shapes and sizes. By using pressure-sensitive tape (Kefu Medical Technology Co., Ltd), the wires were fixed tightly onto the E-slime. The digital multimeter (TH1963A, Tonghui Electronics Co., Ltd) could detect electrical signals in real time.

Cell Viability Tests of E-Slime: C2C12 myoblasts were purchased from ATCC and maintained in the standard CO₂ incubator (Thermo Scientific Heracel 150i) with the environment as 5% CO₂ and 37 °C, while cells were cultured in the growth medium (GM) of Dulbecco's Modified Eagle's Medium as supplemented with 10% Fetal Bovine Serum and 1% Penicillin-Streptomycin-Neomycin (Gibco). Samples of hydrogels, including PVA, PVA/gly, PVA/gly/TA, and E-slime, were washed with 75% ethanol and exposed under UV light for sterilization prior to experiments. Each of the samples (2 g) was immersed in GM for 2 h; after removing the samples, the remaining solutions were used to test for biocompatibility. C2C12 cells were first seeded in Petri dishes at a density of 3.0×10^4 cells cm⁻² and cultured in GM for 24 h. Then, GM was substituted with different remaining solutions for culture cells for another 2 h. In addition, the cells continued to grow in GM as a control. After that, a LIVE/DEAD Cell Imaging kit containing Calcein AM and Propidium Iodide (Life Technologies Co., Ltd) was used to calculate the cell viability.

Ethics Approval Statement: The Human Subjects Ethics Subcommittee of the City University of Hong Kong gave ethical approval for the human test in this work (Internal Reference No. HU-STA-00000215).

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Author Contributions

Y.F. and C.W. contributed equally to this work. Y.F., G.Z., and W.L. conceived the research ideas. Y.F., C.W., and M.C. prepared and fabricated the materials. Y.F., C.W., and H.S. conducted characterizations. Y.F., C.W., M.C., and H.S. conducted motion detection experiments. Y.F., C.W., R.V., W.D., X.Y., G.Z., and W.L. analyzed the experimental data. Y.F., G.Z., and W.L. wrote the manuscript. G.Z. and W.L. supervised the research. All authors discussed the results and reviewed the manuscript.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

electronic slime, epidermal electronics, self-healing sensors, stretchable devices, wearable sensors

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